

# Hand Controller Rewrite

## Expanded Testing Format Draft

This expanded draft is intended for group review before actual execution testing. It keeps the professor's requested structure: detailed test cases, test data, expected results, and coverage for functional and nonfunctional requirements, including valid, invalid, and boundary-value inputs.

Summary: 24 proposed test-data entries, 24 functional test cases, and 8 nonfunctional test cases. Actual result, status, and remarks should be filled in after manual execution.

## Test Data Set

| ID    | Test Data                                                                                             | Category |
|-------|-------------------------------------------------------------------------------------------------------|----------|
| TD-01 | Normal launch environment: Windows machine, .venv ready, camera available, tuning.testing.json loaded | Valid    |
| TD-02 | Selected camera unavailable or disconnected before launch                                             | Invalid  |
| TD-03 | Camera disabled in the control panel before launch                                                    | Invalid  |
| TD-04 | Palm-facing open hand with one clearly visible active hand                                            | Valid    |
| TD-05 | Side-view or visually compressed hand                                                                 | Invalid  |
| TD-06 | No hand visible in frame                                                                              | Invalid  |
| TD-07 | Thumb-index quick pinch and release                                                                   | Valid    |
| TD-08 | Thumb-middle pinch                                                                                    | Valid    |
| TD-09 | Thumb-index hold at about 0.79s, 0.80s, and 0.81s                                                     | Boundary |
| TD-10 | Two left tap cycles at about 1.49s, 1.50s, and 1.51s apart                                            | Boundary |
| TD-11 | ML toggle pose held at about 0.44s, 0.45s, and 0.46s                                                  | Boundary |
| TD-12 | Repeat ML toggle gesture again before cooldown and after cooldown                                     | Boundary |
| TD-13 | ML hold pose while cursor is moving                                                                   | Valid    |
| TD-14 | ML undo and redo poses while target application is active                                             | Valid    |
| TD-15 | Thumb-ring hold at about 0.34s, 0.35s, and 0.36s                                                      | Boundary |
| TD-16 | Finger hover over alphabet keyboard key then thumb-index pinch                                        | Valid    |
| TD-17 | Finger hover over special keys: Shift, Caps, Backspace, Space, Enter, ESC, TAB                        | Valid    |
| TD-18 | Switch between ABC page and 123 or symbols page                                                       | Valid    |
| TD-19 | Two cameras connected; second camera available during runtime                                         | Valid    |
| TD-20 | Refresh camera source list after camera connection or removal                                         | Valid    |

| ID    | Test Data                                                                        | Category      |
|-------|----------------------------------------------------------------------------------|---------------|
| TD-21 | User closes app while worker is running                                          | Valid         |
| TD-22 | Continuous use for 15 to 30 minutes with normal gesture activity                 | Nonfunctional |
| TD-23 | Observe startup feel: minimize timing, overlay appearance, and UI responsiveness | Nonfunctional |
| TD-24 | Rapid control-panel interactions: refresh, launch, stop, relaunch                | Nonfunctional |

## Functional Test Cases

### Startup, Camera, and Base Control

| ID      | Requirement                                      | Steps                                                                                 | Test Data           | Expected Result                                                                               |
|---------|--------------------------------------------------|---------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------|
| TC-F-01 | Application launch                               | Run .\run-tester.ps1 and wait for control panel plus overlay path startup             | TD-01               | App launches without crash and enters live control path                                       |
| TC-F-02 | Camera unavailable before launch                 | Launch app when the selected camera is unavailable                                    | TD-02               | App does not continue silently with a broken camera state; warning or fallback appears        |
| TC-F-03 | Camera disabled before launch                    | Turn off camera from the UI and press launch                                          | TD-03               | App blocks launch and warns the user that camera must be enabled first                        |
| TC-F-04 | Camera refresh list update                       | Press Refresh and observe the camera-source list                                      | TD-20               | Camera list refreshes without crashing and keeps the UI usable                                |
| TC-F-05 | Live camera switch                               | While app is running, change to another connected camera source                       | TD-19               | Worker transitions to the new camera using the app's controlled restart path                  |
| TC-F-06 | Stop while running                               | Launch the app, then press Stop or close the main window                              | TD-21               | Worker stops cleanly and overlay closes without leaving the app in a stuck state              |
| TC-F-07 | Mouse movement allowed only with correct gating  | Show palm-facing active hand and move it; then repeat using no hand or side-view hand | TD-04, TD-05, TD-06 | Cursor moves only when the movement gates pass and remains blocked otherwise                  |
| TC-F-08 | Control disabled prevents active control actions | Turn control off through ML toggle, then try movement and click gestures              | TD-11, TD-12        | Recognition may continue, but control actions remain blocked until control is turned on again |

### Mouse Interaction and Gesture Safety

| ID      | Requirement                   | Steps                                                                               | Test Data    | Expected Result                                                               |
|---------|-------------------------------|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------|
| TC-F-09 | Single left click             | Hover cursor over a clickable target and do a quick thumb-index pinch and release   | TD-07        | Exactly one left click is triggered                                           |
| TC-F-10 | Right click                   | Hover cursor over target and do a thumb-middle pinch                                | TD-08        | Exactly one right click or context menu action is triggered                   |
| TC-F-11 | Double click inside interval  | Perform two quick left tap cycles within the expected interval                      | TD-07, TD-10 | System recognizes a double click                                              |
| TC-F-12 | Double click outside interval | Repeat the two left tap cycles just beyond the allowed boundary                     | TD-10        | System does not falsely report a valid double click when boundary is exceeded |
| TC-F-13 | Drag below threshold          | Hold thumb-index pinch just below the configured drag threshold, then release       | TD-09        | Drag does not start before the hold threshold is met                          |
| TC-F-14 | Drag at or above threshold    | Hold thumb-index pinch at or above threshold, move hand while holding, then release | TD-09        | Drag starts once threshold is met and ends on release                         |

| ID      | Requirement                         | Steps                                                                    | Test Data | Expected Result                                                                     |
|---------|-------------------------------------|--------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------|
| TC-F-15 | Side-view safety for press gestures | Use a side-view hand and attempt left click, right click, and drag start | TD-05     | Press-like actions are blocked or strongly reduced to prevent accidental activation |
| TC-F-16 | No active hand behavior             | Move hand out of frame during mouse mode                                 | TD-06     | No unintended movement or click continues once active hand is gone                  |

## ML Commands and Keyboard Mode

| ID      | Requirement                       | Steps                                                                       | Test Data    | Expected Result                                                                      |
|---------|-----------------------------------|-----------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------|
| TC-F-17 | ML toggle threshold               | Hold the toggle pose below, at, and above the configured threshold          | TD-11        | Below threshold no toggle occurs; at or above threshold control toggles once         |
| TC-F-18 | ML toggle cooldown                | Trigger toggle once, then repeat again during cooldown and after cooldown   | TD-12        | Repeated toggles inside cooldown are blocked; after cooldown a new toggle is allowed |
| TC-F-19 | ML hold clutch behavior           | Activate hold while cursor is moving and while click gestures are attempted | TD-13        | Movement and click actions freeze while hold is active                               |
| TC-F-20 | ML undo and redo in mouse mode    | Trigger undo and redo while app is in mouse mode                            | TD-14        | System sends Ctrl+Z and Ctrl+Y respectively                                          |
| TC-F-21 | ML undo and redo in keyboard mode | Switch to keyboard mode and repeat undo or redo gestures                    | TD-14, TD-15 | Undo and redo should not fire as active commands in keyboard mode                    |
| TC-F-22 | Keyboard mode toggle threshold    | Hold thumb-ring pinch below, at, and above configured threshold             | TD-15        | Below threshold no mode change; at or above threshold app switches mode              |
| TC-F-23 | Alphabet page typing              | Enter keyboard mode, hover over alphabet keys, then pinch to type           | TD-16        | Correct alphabet key is highlighted and typed                                        |
| TC-F-24 | Special keys and page switching   | Test ABC or 123 switch, Shift, Caps, Backspace, Space, Enter, ESC, and TAB  | TD-17, TD-18 | Correct page switching and key outputs occur; keyboard state reflects the action     |

## Nonfunctional Test Cases

| ID       | Requirement                                  | Steps                                                                                             | Test Data                  | Expected Result                                                                      |
|----------|----------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------|
| TC-NF-01 | Startup responsiveness                       | Launch app and observe how quickly the control panel reacts, minimizes, and shows overlay content | TD-23                      | Startup feels responsive and does not show long visible blocking                     |
| TC-NF-02 | UI responsiveness during preflight           | Observe the UI while launch preflight or camera refresh is running                                | TD-23                      | Main window remains usable and visibly communicates pending state                    |
| TC-NF-03 | Runtime stability                            | Run the application continuously for 15 to 30 minutes with normal gesture use                     | TD-22                      | No crash, unrecoverable lag, or stuck overlay appears during sustained use           |
| TC-NF-04 | Reliability under camera fallback conditions | Start with missing camera or change available camera sources before launch or during refresh      | TD-02, TD-19, TD-20        | System falls back gracefully or warns the user instead of silently misbehaving       |
| TC-NF-05 | Controlled worker shutdown                   | Start, stop, and relaunch the app repeatedly from the control panel                               | TD-24                      | Worker thread and overlay stop and start cleanly without leaving broken state behind |
| TC-NF-06 | Consistency and repeatability of gestures    | Repeat the same gesture several times under similar conditions                                    | TD-07, TD-08, TD-13, TD-16 | Results remain consistent enough for normal user operation and do not feel random    |
| TC-NF-07 | Usability of camera source controls          | Use camera source combo box and refresh flow without reading code                                 | TD-19, TD-20               | Controls are understandable enough for a tester to operate with minimal confusion    |

| ID       | Requirement           | Steps                                                                                          | Test Data                  | Expected Result                                                                            |
|----------|-----------------------|------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------|
| TC-NF-08 | Safe failure behavior | Trigger invalid states such as no hand, side-view hand, disabled camera, or unavailable camera | TD-02, TD-03, TD-05, TD-06 | System blocks unsafe actions and fails gracefully instead of generating uncontrolled input |

Note: This expanded version is more complete than the initial draft, but it is still a pre-execution planning document. Actual pass or fail status, observed results, defects, and remarks should be filled in during the group testing activity.